



MAASTRICHT UNIVERSITY
DEPARTMENT OF ADVANCED COMPUTER SCIENCE
DATA SCIENCE FOR DECISION MAKING

KEN4113 - Data Mining Churn Problem of Maastricht Telecom

Author:
Christos Koromilas

Student Number:
i6345703

January 31, 2024

Link to the folder for this assignment:
Google Drive

1 Introduction

In the competitive world of telecommunications, companies are always searching for ways to keep their customers from leaving. They have built big databases with information about all their customers. They use these databases to cut costs and reduce how much money they lose each year. A key part of their strategy is to find out which customers might leave for another company. These customers are known as 'churned customers.' By identifying these customers early, telecommunications companies can work hard to keep them, which helps the company save money and stay strong in a tough market

2 Data Understanding

For this study, we received a dataset comprising churn and non-churn customers from Maastricht Telecommunication for the year 2020. The dataset contains a mix of categorical and numerical features. Numerical features are expressed as integers, whereas categorical features are represented as objects, each with a diverse count of unique values.

The dimensions of the train and test sets are:

- Train set dimensionality: 4695×20
- Test set dimensionality: 2348×20

Both the train and test sets include the following types of columns:

- **int64**: tenure, SeniorCitizen
- **float**: MonthlyCharges
- **object**: gender, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, TotalCharges, Churn

The datasets are complete with no missing values initially. The unique values in each set are summarized in the following table:

Table 1: Unique Values in Train and Test Sets

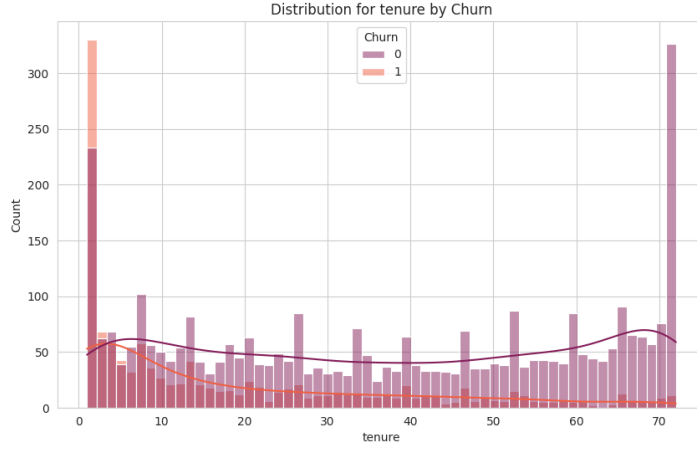
Column	Train Unique Values	Test Unique Values
gender	2	2
SeniorCitizen	2	2
Partner	2	2
Dependents	2	2
tenure	73	73
PhoneService	2	2
MultipleLines	3	3
InternetService	3	3
OnlineSecurity	3	3
OnlineBackup	3	3
DeviceProtection	3	3
TechSupport	3	3
StreamingTV	3	3
StreamingMovies	3	3
Contract	3	3
PaperlessBilling	2	2
PaymentMethod	4	4
MonthlyCharges	1423	1077
TotalCharges	4429	2284
Churn	2	2

It was later discovered that one of the columns, initially identified as an object, was actually of type float. After converting this column to float, some missing values emerged, which were subsequently addressed.

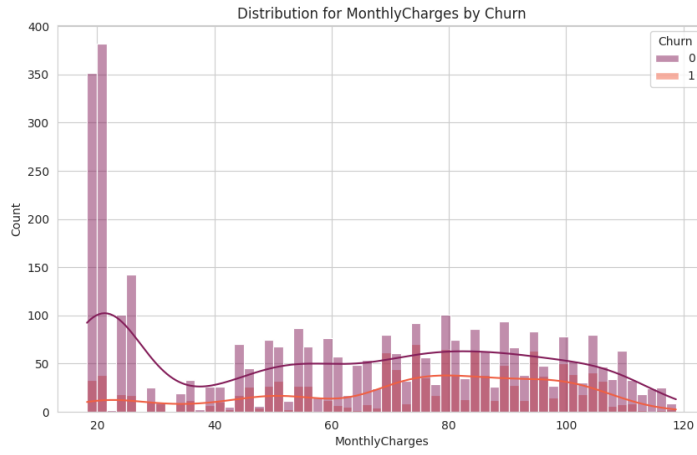
Following this preprocessing, we conducted a numerical analysis (to be detailed in the next section). Post-analysis, we implemented one-hot encoding using the `get_dummies` function from Pandas, transforming each categorical column into multiple binary columns. Consequently, our dataset was entirely numerical, facilitating a comprehensive analysis.

3 Exploratory Data Analysis

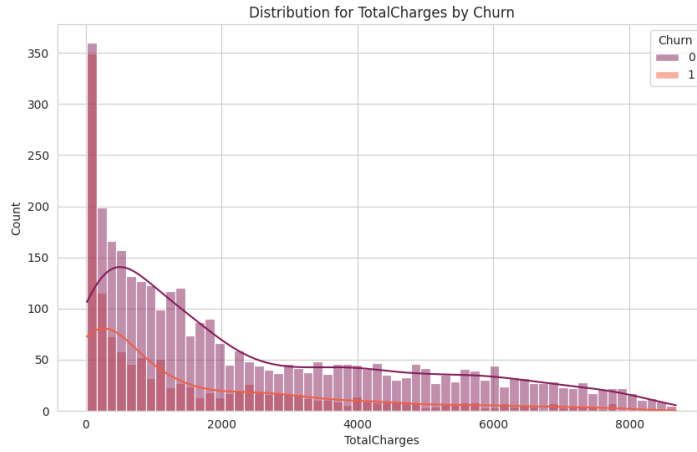
This section presents the distribution of various features within the dataset through histograms. These visualizations provide insights into the data's structure and potential relationships between features and customer churn.



(a) Tenure



(b) Monthly Charges



(c) Total Charges

Figure 1: Distribution of numerical features. (a) Tenure: Shows the distribution of tenure, highlighting the right-skewed nature of customer tenure with the company. (b) Monthly Charges: Indicates most customers have mid to low range monthly charges. (c) Total Charges: Illustrates the overall charges a customer has paid over time, indicative of customer loyalty and the duration of service usage.

The provided histograms display distinct churn patterns across various numerical features. Customers with a shorter tenure appear more prone to churn, suggesting issues with early service experience or competitive market offers. The Monthly Charges histogram indicates higher churn at increased price points, implying that customers are sensitive to pricing. Conversely, the Total Charges histogram shows that customers with higher aggregate payments over time are less likely to churn, which may reflect a developed sense of loyalty or satisfaction with the services rendered. These observations highlight critical areas such as pricing strategies, service quality in the initial stages of customer engagement, and the potential benefits of long-term customer retention programs.



Figure 2: Correlation coefficients of all features with Churn

In the subsequent analysis, after transforming categorical data into numerical format using binary encoding, we assessed the correlations amongst features and with churn. The resulting correlation heatmap and bar graph provide a visual representation of how each feature relates to customer churn. In the heatmap, we can observe clusters of related features due to one-hot encoding, which creates multiple binary variables from a single categorical variable. The intensity of the colors reflects the strength and direction of the relationship, with darker shades indicating stronger correlations.

The bar graph focusing on the correlation with churn gives a more direct insight into which features are most strongly associated with churn. Long tenure and two-year contracts show a strong negative correlation with churn, suggesting that customers who have stayed with the company longer or who have committed to a longer-term contract are less likely to leave. Conversely, month-to-month contracts, lack of online security, absence of tech support, and fiber optic internet services show a positive correlation with churn, indicating higher churn rates for customers with these features. These visualizations are instrumental in highlighting key areas that could influence customer retention strategies.

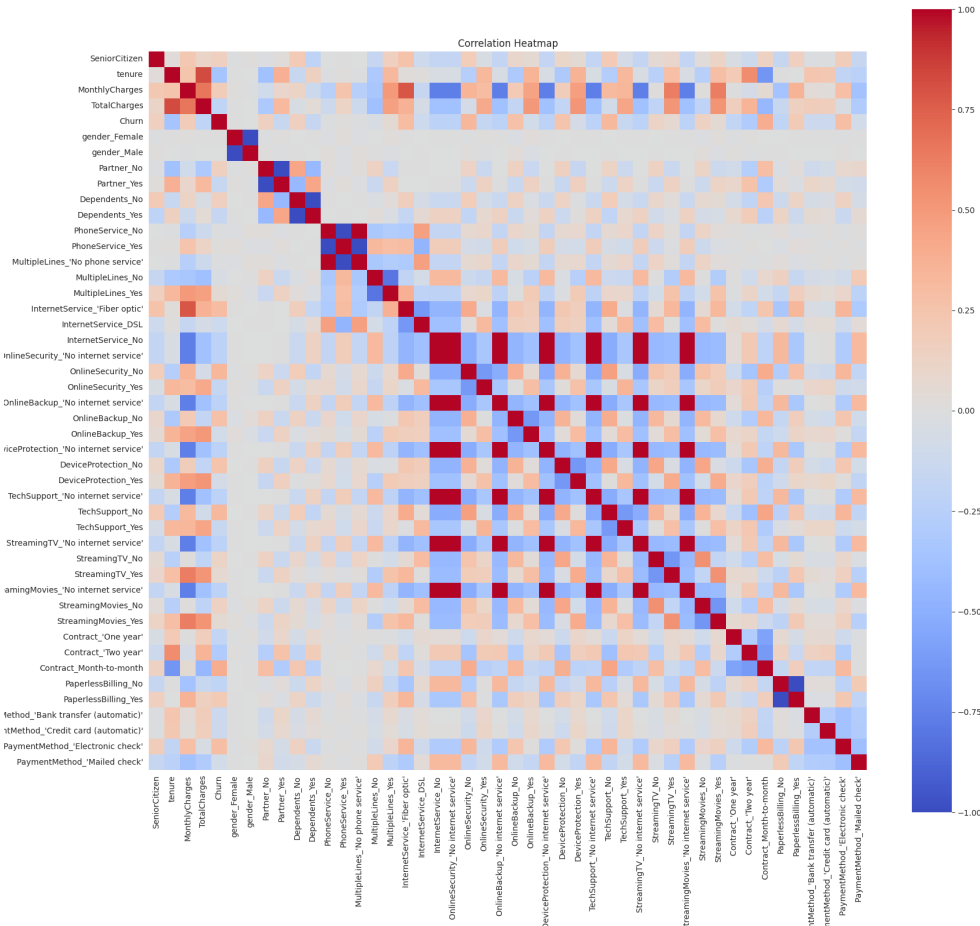
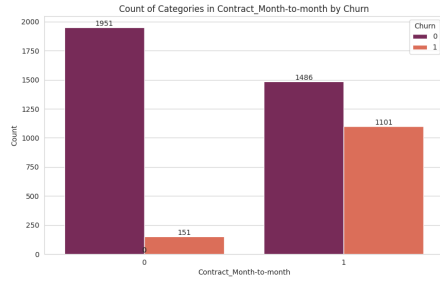
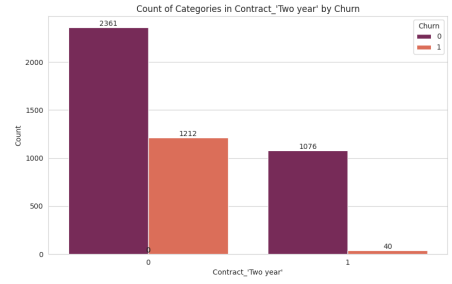


Figure 3: Correlation heatmap between all features

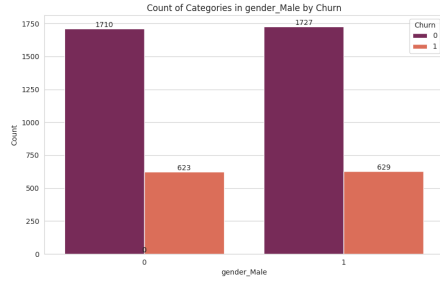
About the categorical columns that we converted as numerical columns with 0s and 1s we took the plots with also the number of 0s and 1s of Churn (No and Yes) in order to see how the most correlated and the non correlated data presenting in the data.



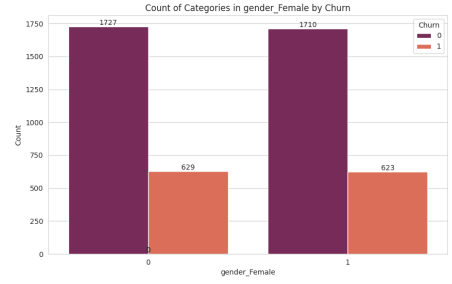
(a) Contrat Month to Month



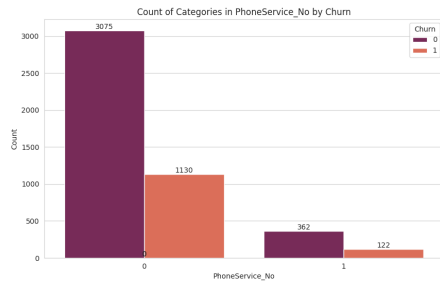
(b) Contract Two Year



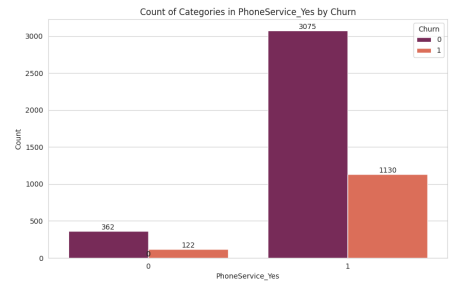
(c) Gender Male



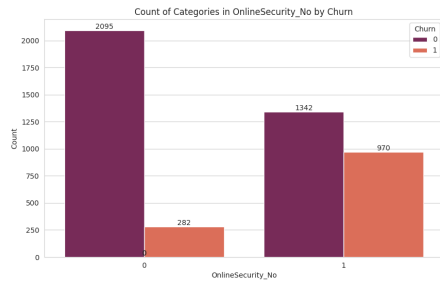
(d) Gender Female



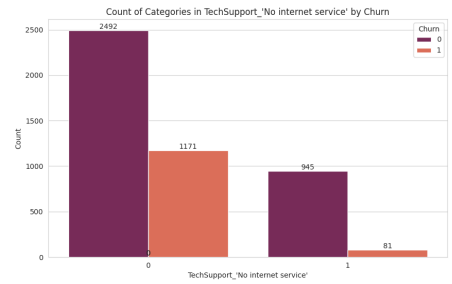
(e) No Phone Service



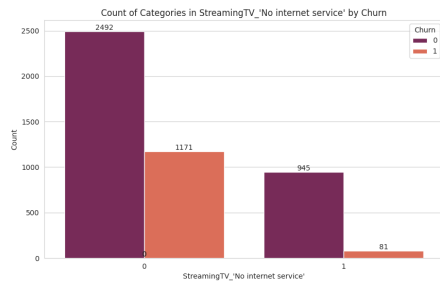
(f) Yes Phone Service



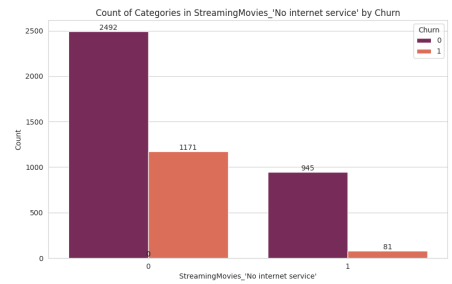
(g) Online Security No



(h) Techsupport - No Internet Service



(i) Streaming TV - No Internet Service



(j) Streaming Movies - No Internet Service

Figure 4: Bargraphs of High and low correlated features with Churn column

The analysis of the churn dataset reveals insightful patterns: customers on month-to-month contracts are more likely to churn, implying a need for strategies to foster longer-term commitments. Gender does not significantly impact churn, suggesting a uniform approach across demographics could be effective. Notably, the absence of internet-based services, such as online security, tech support, and streaming, correlates with higher churn rates, highlighting these as critical areas for improving customer retention.

4 Predictive modelling

4.1 Preparation of the Dataset

This section of the report examines the application of logistic regression and decision tree models to our dataset and evaluates their influence on feature behavior.

Initially, the dataset was standardized to ensure uniformity in the feature scale. This standardized dataset was then split into training and testing sets to validate the model's performance.

4.2 Logistic regression

The logistic regression model was trained with a maximum of 1000 iterations. The resulting feature importance, as determined by the model coefficients, is depicted in the figure 5:

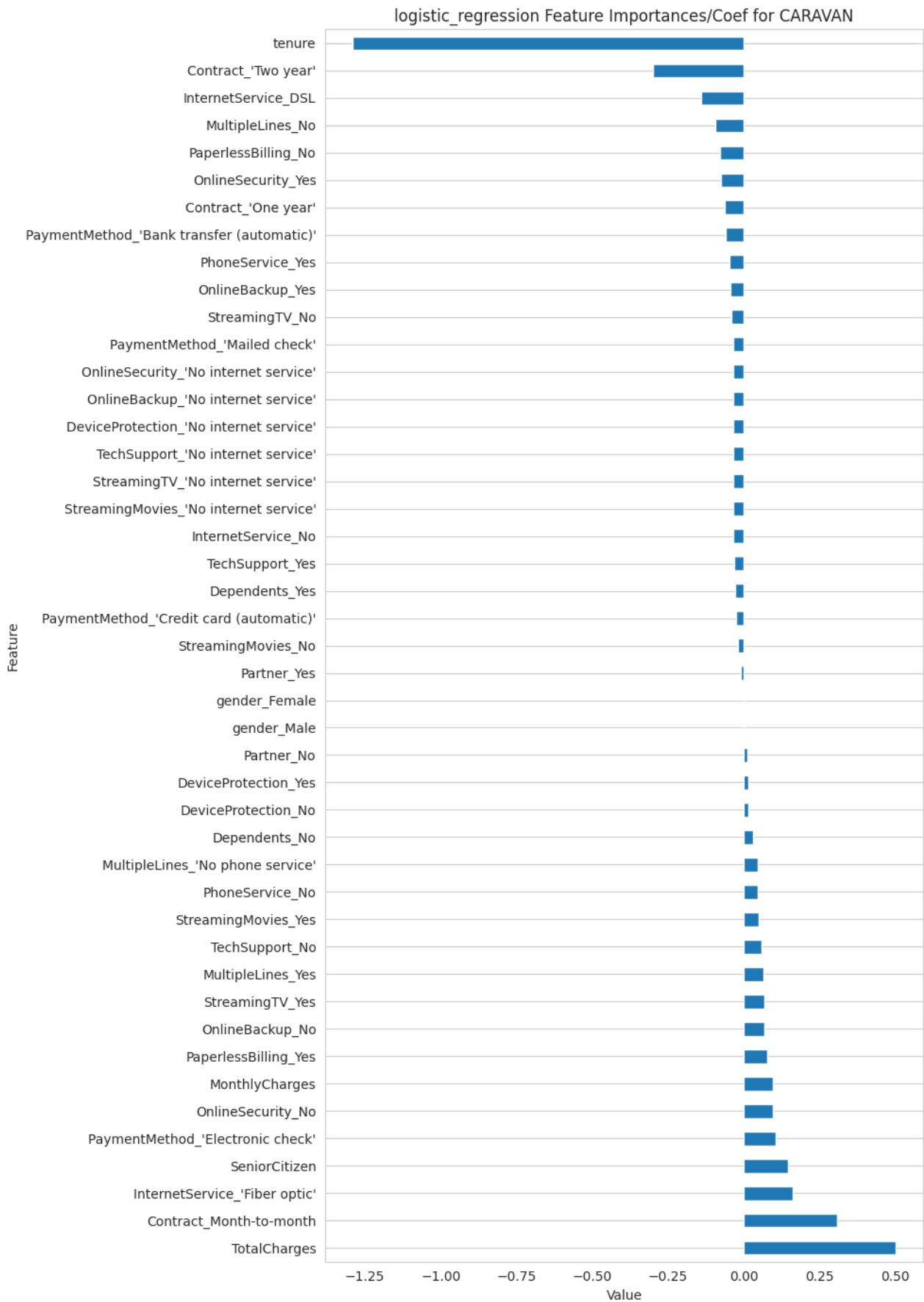


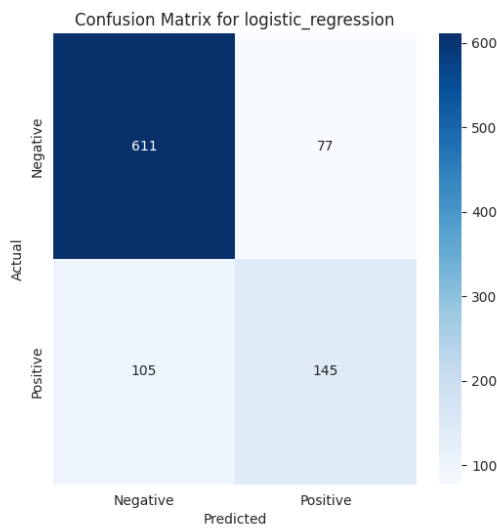
Figure 5: Feature importances from Logistic Regression

Here is the classification report: Subsequently, the confusion matrix and the ROC-AUC

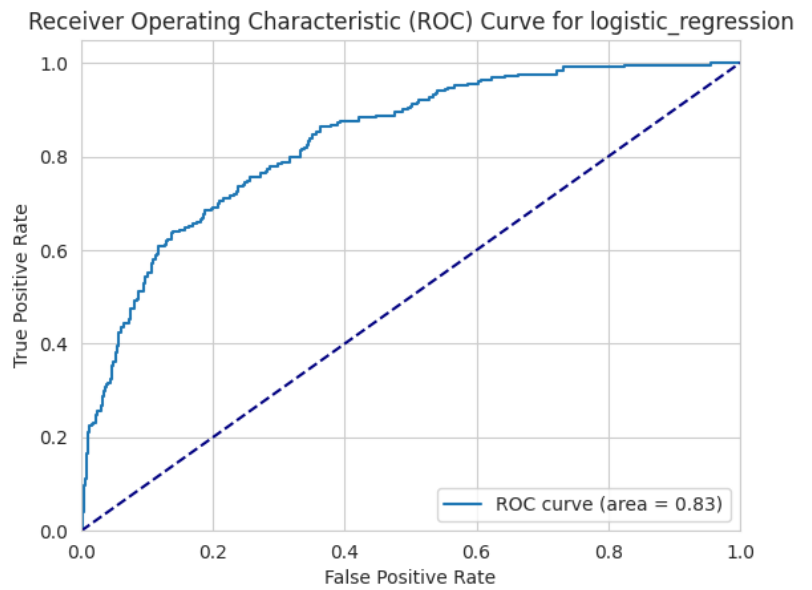
Table 2: Classification Report

	Precision	Recall	F1-Score	Support
0	0.85	0.89	0.87	688
1	0.65	0.58	0.61	250
accuracy			0.81	938
macro avg	0.75	0.73	0.74	938
weighted avg	0.80	0.81	0.80	938

curve were analyzed, as illustrated below:



(a) The confusion matrix for the Logistic Regression



(b) The ROC-AUC curve for the Logistic Regression

4.3 Decision Tree

The decision tree model's feature importance is visualized in the figure below:

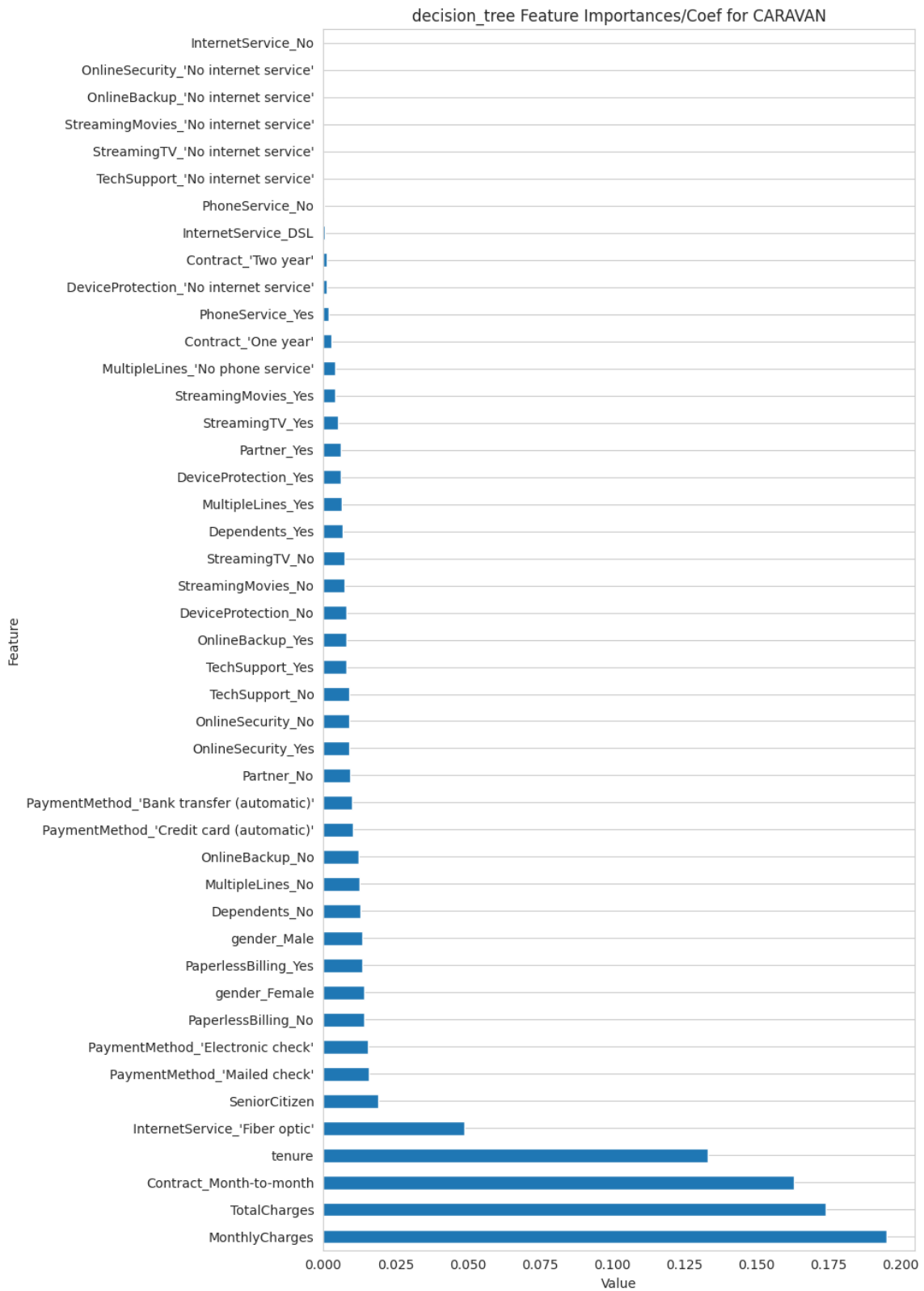


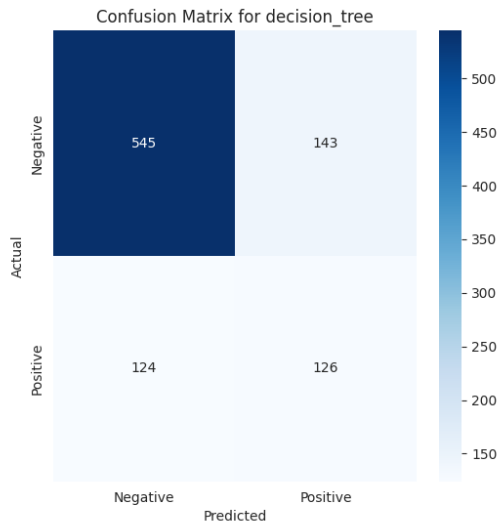
Figure 7: Feature importances from Decision Tree

The classification report is as follows:

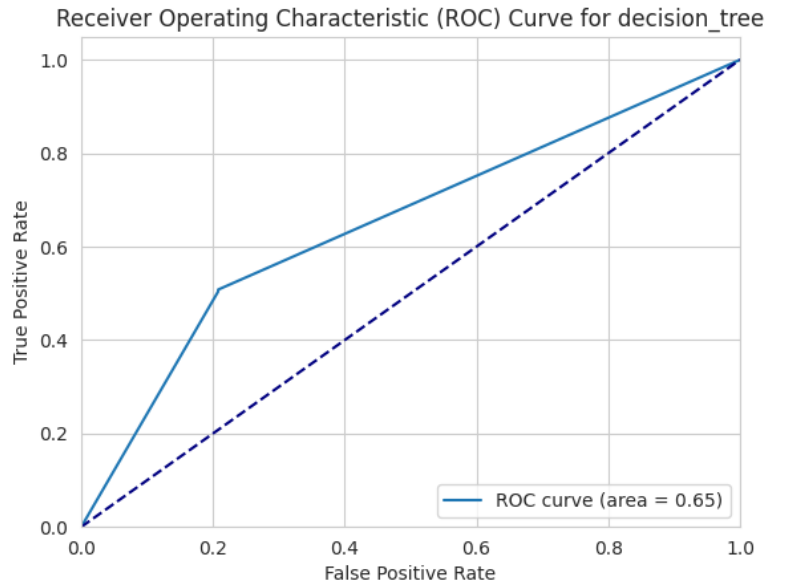
Table 3: Classification Report

	Precision	Recall	F1-Score	Support
0	0.81	0.79	0.80	688
1	0.47	0.50	0.49	250
accuracy			0.72	938
macro avg	0.64	0.65	0.64	938
weighted avg	0.72	0.72	0.72	938

The confusion matrix and the ROC-AUC curve for the decision tree model are presented below:



(a) The confusion matrix for the Decision Tree



(b) The ROC-AUC curve for the Decision Tree

Bellow you can see also how the tree looks:

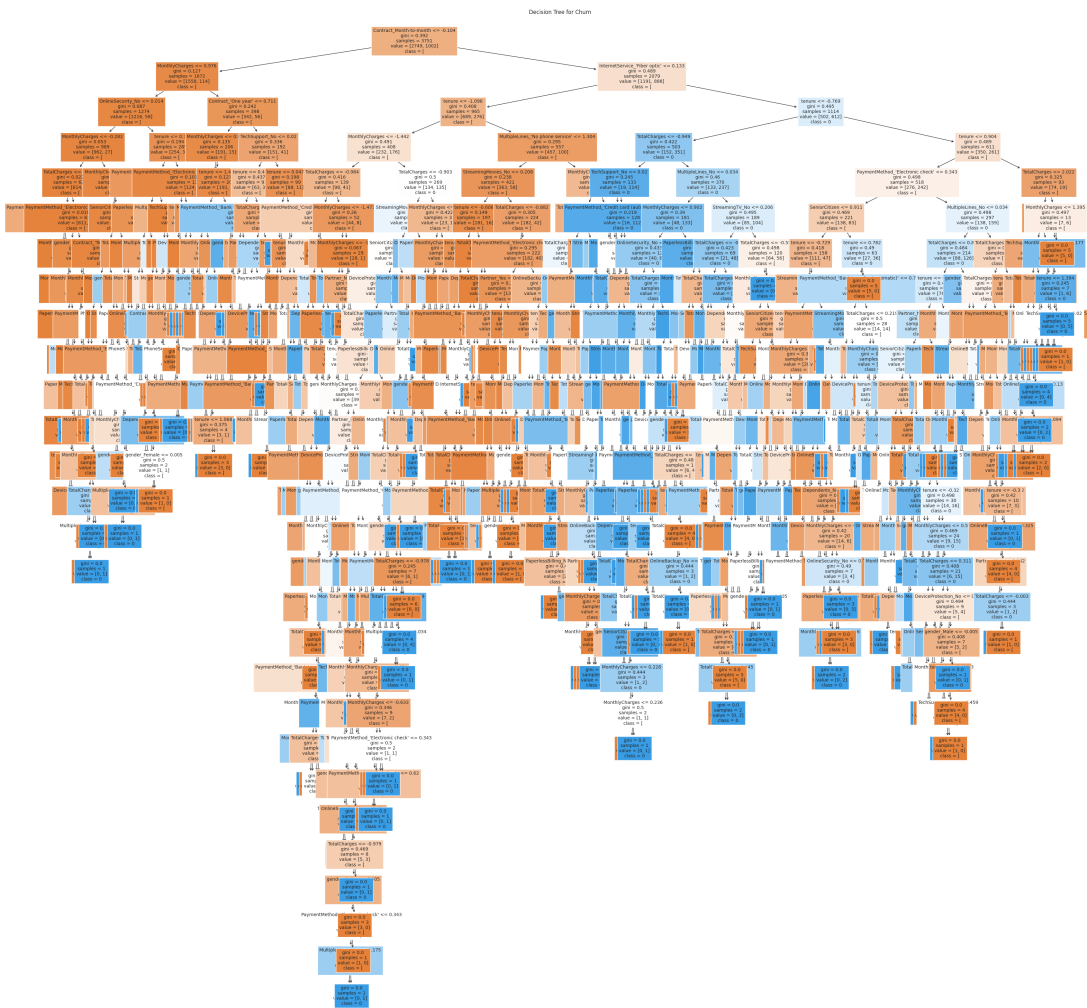


Figure 9: The Decision Tree

5 Improved models

The main problem of the previous models is that we trained them with imbalanced data, with all the features and without the optimum parameters for better performance. So in order to achieve class balancing, feature selection and cross-validation of the model we created a pipeline.

5.1 Pipeline

The pipeline created because its better to organise the whole steps of each combination of the models.

5.1.1 Class balancing

The problem with the unbalanced data we tried to solve it using two different techniques. Undersampling and oversampling of the dataset. For the undersampling we used the Random Under-Sampling and for the oversampling we used the Synthetic Minority Over-Sampling technique (SMOTE). These both methods will help to improve my final model.

5.1.2 Scaler

The Logistic Regression model is very sensitive to the range of data. So we used different ways to scale the data (and also the choice to not scale them) StandardScaler and MinMaxScaler, that the first one is removing the mean and scaling to unit variance, and the second one normalizing the features to the given range.

5.1.3 Feature Selection

For the feature selection part we used two strategies. SelektkBest with chi-squared(ch2) statistical test for non-negative features, and RFE (recursive Feature Elimination). These methos are used to find and choose the most relevant features. Both are helping to improve the explanation of models and parallel reducing the oferfitting.

5.1.4 Classifiers

We used two classifiers as in the previous step, Logistic Regression and Decision Tree. These classifiers will help me get the best features and then test the data to tha model.

5.1.5 Cross-Validation

We used for cross-validation the Stratified K-Fold cross-validation with 5 splits. This crossvalidation gives a more accurate performance estimate for new data and maintains the sample rate for each class.

5.1.6 Pipeline Search

My first idea was to use the known GridSearchCV of sklearn. But we used the Randomized Search because the total combinations of the search is 648. It will took much time. So we chose with randomized search to run the code for 100 of the total combinations.

5.1.7 Scoring

In order to understand which of the metrics is better to explain our model, we used different metrics in order to get the optimum parameters for the pipeline. Metrics: accuracy, AUC, F1, precision and recall. For each metric, we identified the best estimator, fine-tuned the parameters, analyzed the most influential features, and studied the learning curves for deeper insights.

5.2 Results for each metric

The optimization of the machine learning pipeline was conducted with a focus on five key metrics: accuracy, AUC, F1, precision, and recall. The pipeline was executed multiple times, each time using a different metric as the optimization criterion. This approach allowed for a comprehensive understanding of how the model performs across various aspects of classification tasks. Below are the results and discussions for each metric:

5.2.1 Accuracy

Optimizing for accuracy yielded a model that correctly predicted 78.57% of the outcomes. This model is robust for general purposes where the costs of false positives and false negatives are similar.

Metric	Value
Accuracy	0.7857
Precision	0.5911
Recall	0.6360
F1 Score	0.6127
AUC	0.8213

Table 4: Performance Metrics

For the accuracy the learning curve:

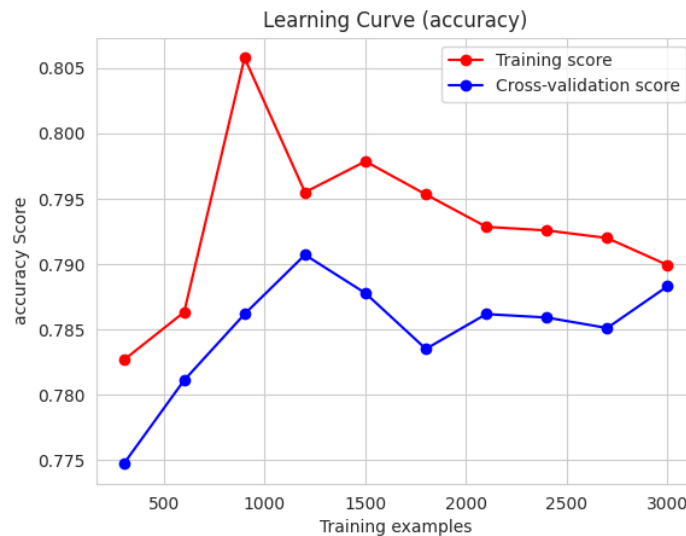
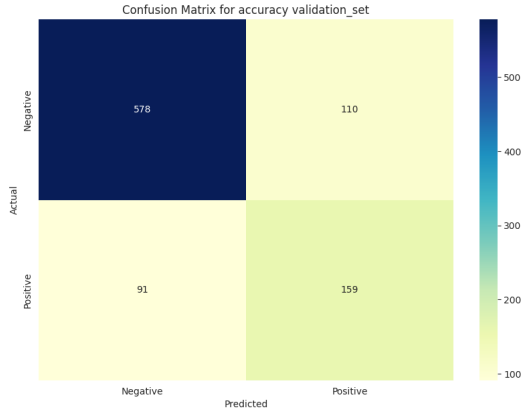
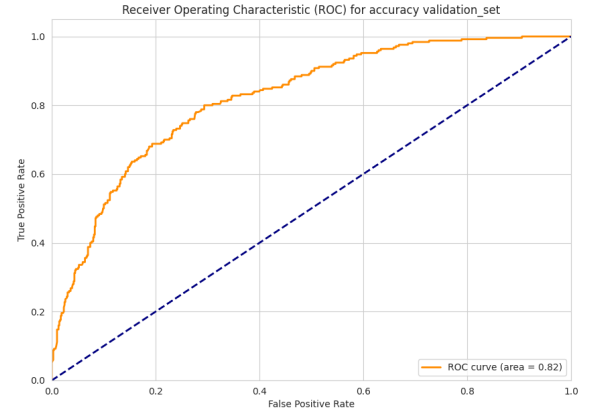


Figure 10: Learning Curve for Accuracy

The confusion matrix and the ROC-AUC curve:



(a) Confusion Matrix for Accuracy



(b) ROC-AUC Curve for Accuracy

5.2.2 AUC

The AUC optimization resulted in a model with a score of 0.8323, which reflects its excellent ability to differentiate between the positive and negative classes. This model is particularly useful in cases where distinguishing between classes is more important than the exact prediction.

Metric	Value
Accuracy	0.7367
Precision	0.5039
Recall	0.7720
F1 Score	0.6098
AUC	0.8323

Table 5: Performance Metrics

For the AUC the learning curve:

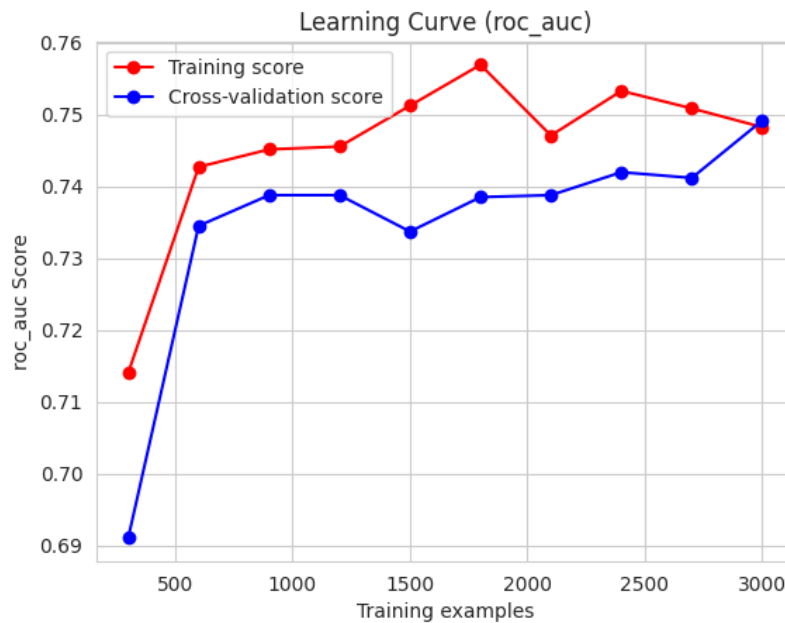
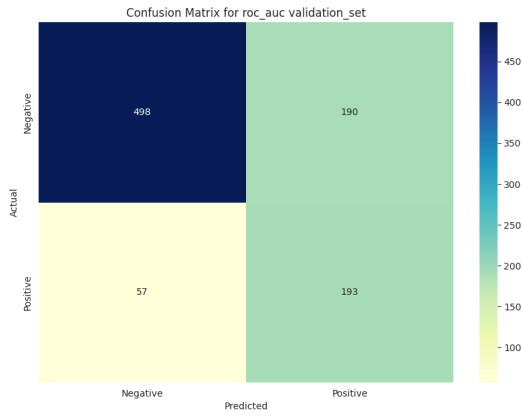
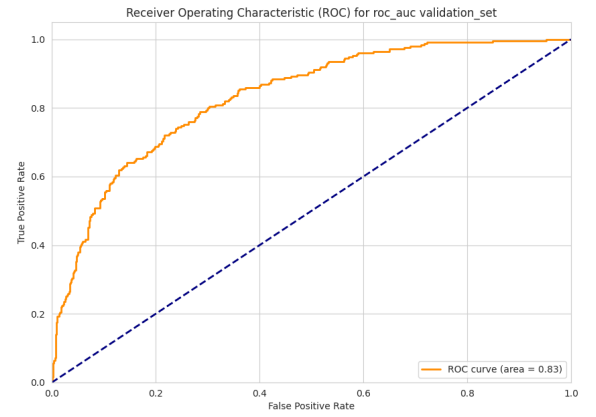


Figure 12: Learning Curve for ROC AUC

The confusion matrix and the ROC-AUC curve:



(a) Confusion Matrix for ROC AUC



(b) ROC-AUC Curve for ROC AUC

5.2.3 F1

The F1-optimized model achieved a score of 0.6164, balancing precision and recall. This model is best applied in situations where false positives and false negatives have a similar cost, and a balance between them is required.

Metric	Value
Accuracy	0.7399
Precision	0.5078
Recall	0.7840
F1 Score	0.6164
AUC	0.8326

Table 6: Performance Metrics

For the F1 the learning curve:

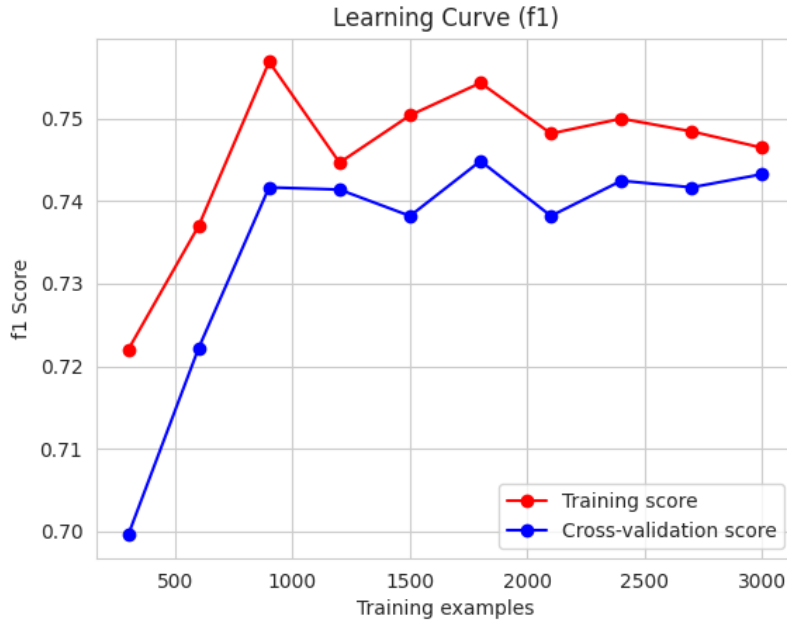
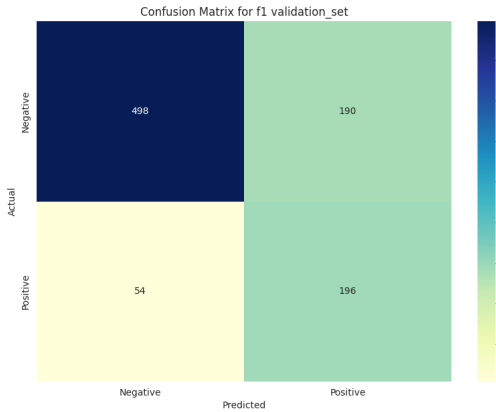
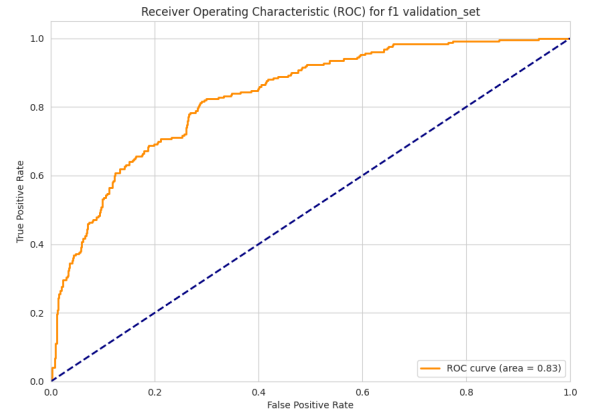


Figure 14: Learning Curve for F1

The confusion matrix and the ROC-AUC curve:



(a) Confusion Matrix for F1



(b) ROC-AUC Curve for F1

5.2.4 Precision

With a precision focus, the model achieved a score of 0.6296, indicating a strong ability to minimize false positives. This is essential in scenarios where the cost of a false positive is much higher than a false negative.

Metric	Value
Accuracy	0.7932
Precision	0.6296
Recall	0.5440
F1 Score	0.5837
AUC	0.8206

Table 7: Performance Metrics

For the precision the learning curve:

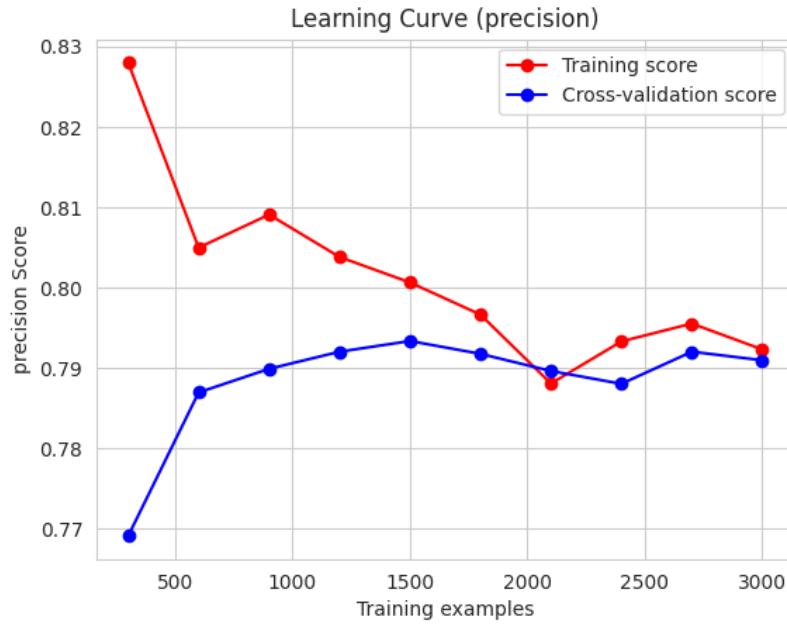
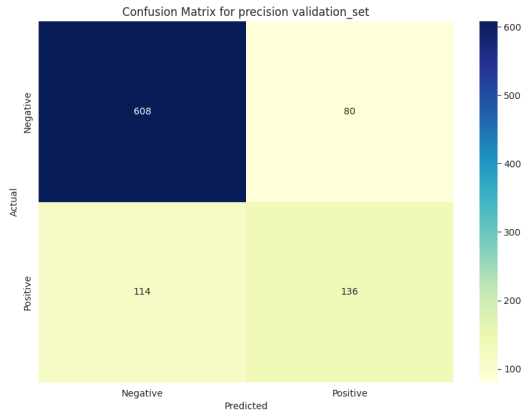
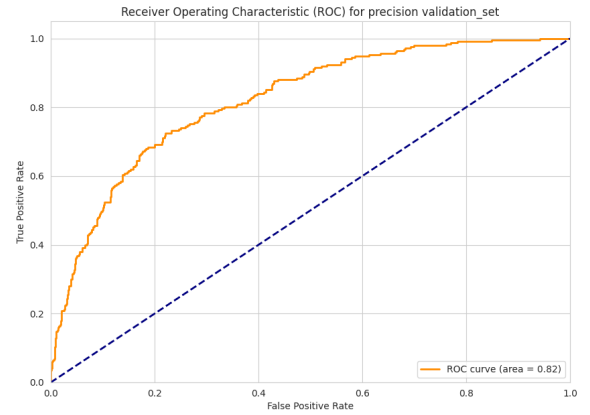


Figure 16: Learning Curve for Precision

The confusion matrix and the ROC-AUC curve:



(a) Confusion Matrix for Precision



(b) ROC-AUC Curve for Precision

5.2.5 Recall

The recall-optimized model achieved a score of 0.8000, indicating it is highly capable of identifying all relevant instances. This model is critical in situations where missing a positive prediction has a higher cost compared to making a false positive error.

Metric	Value
Accuracy	0.6919
Precision	0.4556
Recall	0.8000
F1 Score	0.5806
AUC	0.7524

Table 8: Performance Metrics

For the recall the learning curve:

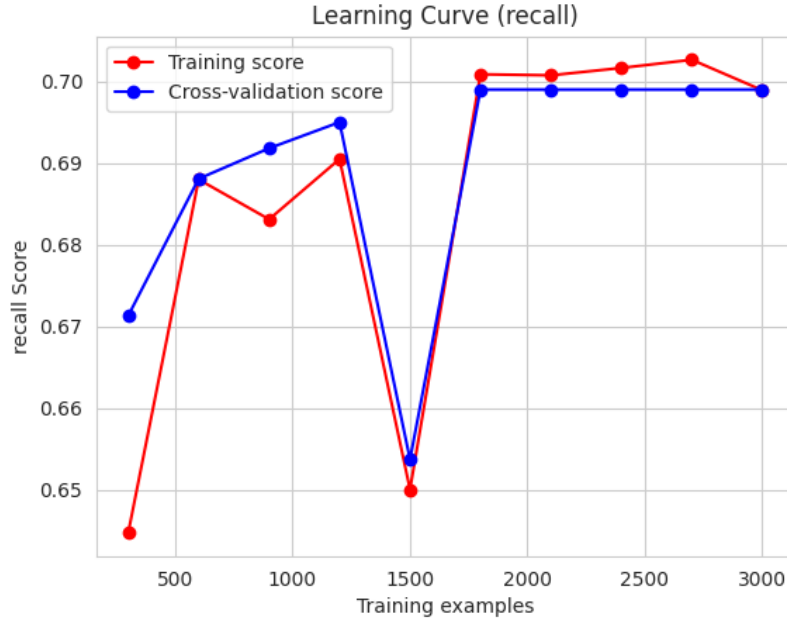
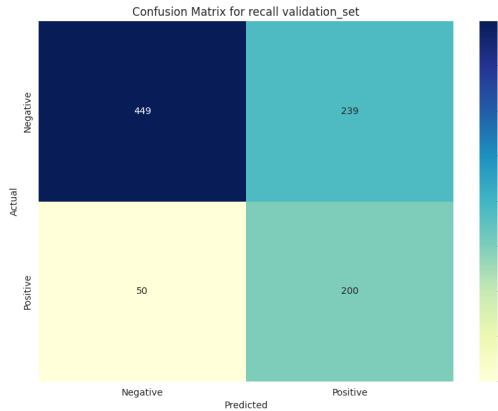
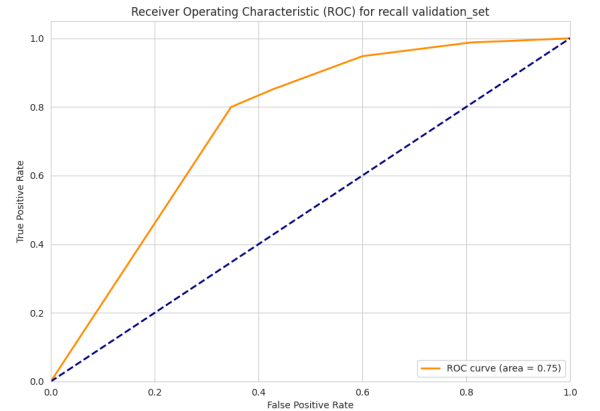


Figure 18: Learning Curve for Recall

The confusion matrix and the ROC-AUC curve:



(a) Confusion Matrix for Recall



(b) ROC-AUC Curve for Recall

Each of these metrics provides a different perspective on the model's performance and can guide the selection of the most appropriate model for a given application. The optimization process involved fine-tuning hyperparameters, feature selection, and balancing the dataset to

ensure that the model's performance is not only high but also consistent across different evaluation metrics.

6 Explain the Features

Table 9: Important Features for Churn Analysis

Feature Name	Count
tenure	4
PaymentMethod_‘Mailed check’	4
Partner_Yes	4
PaperlessBilling_No	4
Contract_Month-to-month	4
Contract_‘Two year’	4
StreamingTV_‘No internet service’	4
TechSupport_‘No internet service’	4
TechSupport_No	3
PaymentMethod_‘Electronic check’	3
StreamingTV_Yes	3
TotalCharges	3
PaymentMethod_‘Credit card (automatic)’	3
StreamingMovies_‘No internet service’	3
PaymentMethod_‘Bank transfer (automatic)’	3
DeviceProtection_‘No internet service’	3
OnlineBackup_Yes	3
OnlineBackup_‘No internet service’	3
OnlineSecurity_Yes	3
Dependents_Yes	3
InternetService_No	3
TechSupport_Yes	2
DeviceProtection_Yes	2
DeviceProtection_No	2
Dependents_No	2
PaperlessBilling_Yes	2
StreamingMovies_Yes	2
InternetService_‘Fiber optic’	2
StreamingTV_No	2
OnlineBackup_No	2
MultipleLines_No	2
InternetService_DSL	2
MonthlyCharges	2
SeniorCitizen	2
OnlineSecurity_‘No internet service’	2
OnlineSecurity_No	2
Contract_‘One year’	2
PhoneService_No	1
PhoneService_Yes	1
MultipleLines_‘No phone service’	1
gender_Female	1
gender_Male	1
Partner_No	1
StreamingMovies_No	1

This analysis clearly indicates that certain features, such as 'tenure', various 'PaymentMethods', and 'Contract' types, play pivotal roles in customer churn. Focusing on these areas could yield substantial improvements in customer retention strategies.

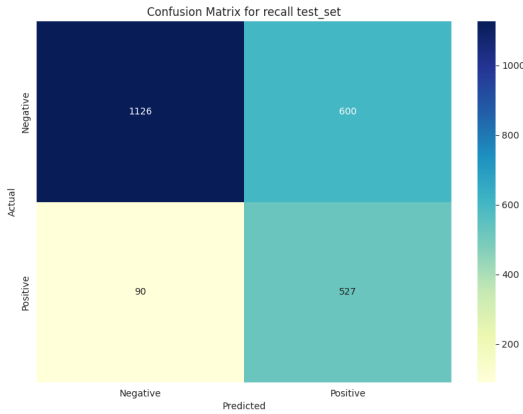
7 Predict the 300 customers and the cost

Using the model optimized for recall, we evaluated its performance on a test set. The confusion matrix and ROC-AUC curve for the test set are shown in Figure 20a and Figure 20b, respectively. The metrics on table10 while using the test set:

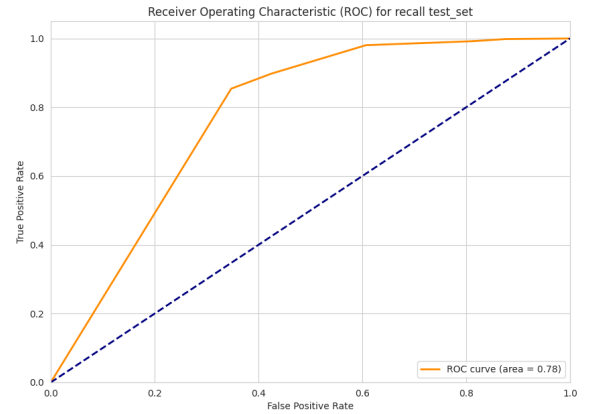
Metric	Value
Accuracy	0.7055
Precision	0.4676
Recall	0.8541
F1 Score	0.6044
AUC	0.7769

Table 10: Performance Metrics

The confusion matrix and the ROC-AUC curve for test set:



(a) Confusion Matrix for Test-Set



(b) ROC-AUC Curve for Test-Set

7.1 Selection of Churn Customers

From the test set, we selected 300 customers that the model predicted as likely to churn. This selection was based on the model's ability to maximize the identification of actual churn cases, as the recall metric is the most critical in this scenario. Out of these 300 customers, 141 were true churn customers.

7.2 Cost Calculation for the Predictive Model

To evaluate the financial impact of the model's predictions, we conducted a cost analysis for the entire test set under the following conditions:

- Every customer predicted as churn, correctly or incorrectly, would receive a gift valued at 10 euros.
- Each actual churn customer incorrectly predicted as non-churning (false negative) would result in a loss equivalent to the average monthly subscription fee of 64 euros.

The confusion matrix 20b from the model's performance on the test set provided the necessary figures for this calculation:

- True Positives (TP): 527
- False Positives (FP): 600
- False Negatives (FN): 90

Using these values, the total expected cost is calculated by summing the cost for all predicted churns (both TP and FP) and the losses from FN:

$$\text{Total Cost} = (\text{TP} + \text{FP}) \times 10 + \text{FN} \times 64$$

With the given numbers, the expected cost for Maastricht Telecom when using this model on the test set is:

$$\text{Total Cost} = (527 + 600) \times 10 + 90 \times 64 = 17,030 \text{ euros}$$

This cost reflects the financial implications of applying the churn prediction model in a real-world scenario and emphasizes the importance of precision in predicting churn to minimize unnecessary expenses.

Table 11: Confusion Matrix for the Test Set

	Predicted	
	Non-Churn	Churn
Actual Non-Churn	1126	600
Actual Churn	90	527

Appendix: Best Parameters and Features for Models Optimizing for Accuracy

Table 12: Best Parameters for Accuracy

Parameter	Value
selector__k	25
selector	SelectKBest(k=25)
scaler	MinMaxScaler()
resampler	SMOTE()
classifier__solver	'lbfgs'
classifier__penalty	'l2'
classifier__C	100
classifier	LogisticRegression(C=100)

Table 13: Selected Features for the Best Accuracy Model

Selected Features
tenure
Partner_Yes
Dependents_Yes
InternetService_'Fiber optic'
InternetService_DSL
InternetService_No
OnlineSecurity_'No internet service'
OnlineSecurity_No
OnlineSecurity_Yes
OnlineBackup_'No internet service'
OnlineBackup_Yes
DeviceProtection_'No internet service'
DeviceProtection_Yes
TechSupport_'No internet service'
TechSupport_No
TechSupport_Yes
StreamingTV_'No internet service'
StreamingMovies_'No internet service'
Contract_'One year'
Contract_'Two year'
Contract_Month-to-month
PaperlessBilling_No
PaymentMethod_'Bank transfer (automatic)'
PaymentMethod_'Credit card (automatic)'
PaymentMethod_'Mailed check'

Optimizing for AUC

Table 14: Best Parameters for AUC

Parameter	Value
selector__n_features_to_select	30
selector	RFE(estimator=LogisticRegression(C=100), n_features_to_select=30)
scaler	MinMaxScaler()
resampler	RandomUnderSampler()
classifier__solver	'saga'
classifier__penalty	'l2'
classifier__C	100
classifier	LogisticRegression(C=100, solver='saga')

Table 15: Selected Features for the Best AUC Model

Selected Features
SeniorCitizen
tenure
MonthlyCharges
TotalCharges
Partner_Yes
Dependents_Yes
MultipleLines_No
InternetService_'Fiber optic'
InternetService_DSL
InternetService_No
OnlineSecurity_No
OnlineSecurity_Yes
OnlineBackup_'No internet service'
OnlineBackup_No
DeviceProtection_'No internet service'
TechSupport_'No internet service'
TechSupport_No
StreamingTV_'No internet service'
StreamingTV_No
StreamingTV_Yes
StreamingMovies_'No internet service'
StreamingMovies_Yes
Contract_'One year'
Contract_'Two year'
Contract_Month-to-month
PaperlessBilling_No
PaperlessBilling_Yes
PaymentMethod_'Bank transfer (automatic)'
PaymentMethod_'Electronic check'
PaymentMethod_'Mailed check'

Optimizing for F1

Table 16: Best Parameters for F1 Score

Parameter	Value
selector__n_features_to_select	20
selector	RFE(estimator=LogisticRegression(C=10, solver='liblinear'), n_features_to_select=20)
scaler	MinMaxScaler()
resampler	RandomUnderSampler()
classifier__solver	'liblinear'
classifier__penalty	'l2'
classifier__C	10
classifier	LogisticRegression(C=10, solver='liblinear')

Table 17: Selected Features for the Best F1 Score Model

Selected Features
SeniorCitizen
tenure
MonthlyCharges
TotalCharges
Partner_Yes
Dependents_No
PhoneService_No
PhoneService_Yes
MultipleLines_'No phone service'
MultipleLines_No
OnlineSecurity_Yes
OnlineBackup_Yes
DeviceProtection_No
StreamingTV_Yes
Contract_'Two year'
Contract_Month-to-month
PaperlessBilling_No
PaymentMethod_'Credit card (automatic)'
PaymentMethod_'Electronic check'
PaymentMethod_'Mailed check'

Optimizing for Precision

Table 18: Best Parameters for Precision

Parameter	Value
selector__n_features_to_select	30
selector	RFE(estimator=LogisticRegression(C=100), n_features_to_select=30)
scaler	MinMaxScaler()
resampler	SMOTE()
classifier__solver	'lbfgs'
classifier__penalty	'l2'
classifier__C	100
classifier	LogisticRegression(C=100)

Table 19: Selected Features for the Best Precision Model

Selected Features
tenure
TotalCharges
gender_Female
gender_Male
Partner_No
Partner_Yes
Dependents_No
Dependents_Yes
InternetService_No
OnlineSecurity_'No internet service'
OnlineBackup_'No internet service'
OnlineBackup_No
OnlineBackup_Yes
DeviceProtection_'No internet service'
DeviceProtection_No
DeviceProtection_Yes
TechSupport_'No internet service'
TechSupport_No
TechSupport_Yes
StreamingTV_'No internet service'
StreamingTV_No
StreamingTV_Yes
StreamingMovies_No
StreamingMovies_Yes
PaperlessBilling_No
PaperlessBilling_Yes
PaymentMethod_'Bank transfer (automatic)'
PaymentMethod_'Credit card (automatic)'
PaymentMethod_'Electronic check'
PaymentMethod_'Mailed check'

Optimizing for Recall

Table 20: Best Parameters for Recall

Parameter	Value
selector__k	5
selector	SelectKBest(k=5)
scaler	MinMaxScaler()
resampler	RandomUnderSampler()
classifier__solver	'saga'
classifier__penalty	'l2'
classifier__C	0.1
classifier	LogisticRegression(C=0.1, solver='saga')

Table 21: Selected Features for the Best Recall Model

Selected Features
TechSupport_'No internet service'
StreamingTV_'No internet service'
StreamingMovies_'No internet service'
Contract_'Two year'
Contract_Month-to-month

Declarations

- I, **Christos Koromilas**, hereby declare that I have not used large language models or any automated tools for generating the written answers and interpretations in this Analytical Report.
- I, **Christos Koromilas**, hereby declare that I have used: large language model to generate the structure and format of this template in $\text{\LaTeX}$.